



IoT-enabled Bus Ticketing Device Project Proposal

Team Name	Syntech-IoT
Category	University Category

1) Problem Definition

1.1) Introduction

Sri Lanka's public bus system is essential for daily commutes but faces persistent challenges such as overcrowding, unreliable schedules, financial mismanagement, and safety concerns. These issues result in commuter discomfort, wasted time, and financial losses for bus operators. The primary cause is the lack of real-time information on bus occupancy, location, and arrival times, forcing operators to rely on **inefficient fixed schedules** due to limited data on ridership patterns.

Additionally, poor compliance with speed limits and route adherence compromises safety and reliability, often going unchecked due to inadequate monitoring.

To address these challenges, our team proposes an **IoT-enabled ticketing machine** that not only streamlines ticketing but also collects real-time operational data. Since ticketing machines are already a mandatory requirement in the public transport sector, integrating IoT functionality into them is a more efficient and cost-effective approach than deploying a separate monitoring system.

This solution will benefit commuters, operators, bus owners, and regulators by improving efficiency, comfort, safety, and decision-making. Moreover, while existing ticketing machines are priced above **Rs. 35,000**, our product offers these advanced features at a significantly lower cost, making it an affordable and effective solution for transport management and network monitoring.

1.2) Problem Analysis

It is very evident that the Sri Lanka's public bus system have become inefficient and unreliable according to public opinion and first-hand experiences, leading to several challenges:

- **Time Wastage:** Unreliable schedules force passengers to wait long periods without knowing when a bus will arrive.

- **Inefficient Route Mapping:** Uneven bus distribution causes overcrowding on some routes and underutilization on others.
- **Limited Night-time Service:** Few buses operate at night, leaving passengers stranded.
- **Lack of Accountability:** No mechanism exists for passengers to report misconduct or poor service.
- **Unsafe Rides:** Reckless driving and poor safety enforcement create hazardous travel conditions.
- **Route Uncertainty:** No reliable system exists to guide commuters, tourists on the best routes to their destination, bus stops for the routes in non-urban areas.
- **Cash Payment Issues:** Dependence on cash leads to overpaying or difficulties in finding exact change and there's no transparency between the operators, owners and the authorities.
- **Environmental Impact:** Paper tickets contribute to litter and waste; an e-ticketing system can reduce this burden.

Our solution wishes to address each of this issues by utilizing our IoT device and its software counterpart.

2)Proposed Solution

2.1) Proposed Product

IoT-enabled Ticket Issuing Machine with mobile transit app

Our device enhances Sri Lanka's public bus network by tracking real-time data and improving efficiency.

Key Features:

- Tracks bus location, occupancy, and speed in real-time.
- Transmits data to a central system for analysis and optimization.
- Display the route data and collect feedback through a mobile application.

Breakdown

1. Real-Time Data Tracking

- **Location Tracking:** GPS provides live bus locations via ticket issuing machine.
- **Speed Monitoring:** Detects speeding; sends alerts for safety.
- **Route Adherence:** Ensures compliance with predefined routes.

2. Ticket Issuing with IoT Integration

- **Passenger Count:** Monitors number of tickets issued.
- **Occupancy Monitoring:** Prevents overcrowding with real-time capacity updates.
- **Payment Options:** Supports cash, contactless, and digital payments.

3. Centralized System & Analysis

- Collects and securely transmits data (location, speed, occupancy).
- Uses AI/ML to identify patterns for route optimization and demand management.

4. Stakeholder Benefits

- **Commuters:** Access live bus updates via an app to reduce waiting times.
- **Operators:** Optimize scheduling and identify inefficiencies.
- **Authorities:** Enforce safety, improve planning, and manage routes better.

5. Notifications and Alerts

- **Commuters:** Get updates on arrivals, seat availability, and delays.
- **Operators/Regulators :** Receive alerts for overcrowding, speeding, and route deviations.

6. Future Scalability

- **Expansion:** Begin with pilot routes and scale to the full network.
- **Integration:** Potential to connect with smart city systems and traffic management.

2.2) Uniqueness of the Solution

1. Upgrade to Existing System

Enhances traditional handheld ticketing machines simplifying adoption by leveraging familiar tools, making implementation easier compared to entirely new systems.

2. Real-Time Speed and Safety Monitoring

Combines GPS and speed tracking to detect reckless driving, proactively ensuring passenger safety and addressing Sri Lanka's road conditions.

3. Passenger Feedback System

Empowers passengers to report issues or commend service, promoting accountability and service improvements beyond just operational metrics.

4. Overloading Control with Ticket Monitoring

Integrates passenger limits and pre-booking to manage passenger capacity, directly addressing overloading and improving boarding efficiency.

5. Real-Time Bus Tracking for Passengers

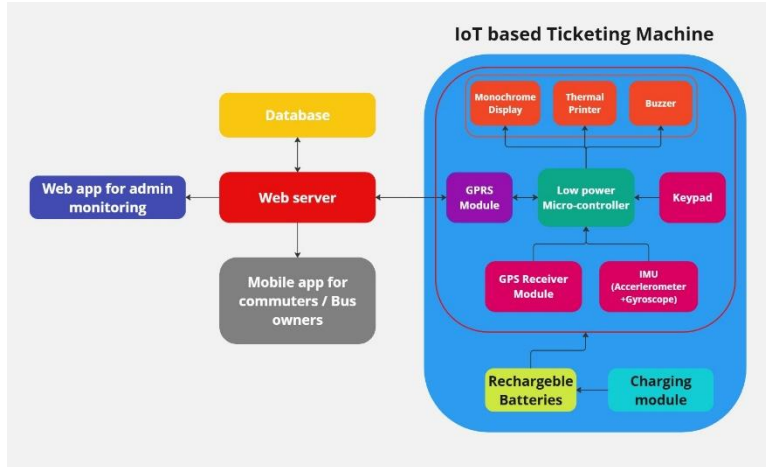
Provides live bus locations and arrival times, a first for Sri Lanka's private and SLTB buses, enhancing trip planning, reliability, and convenience.

6. Data-Driven Insights

Collects actionable data for route optimization, road safety and policy planning, supporting both operational efficiency and long-term infrastructure improvements.

3) Technical Overview and Implementation

3.1) Technical Details



The IoT-enabled ticketing machine will be employing a low-power micro-controller as it needs to operate for a maximum duration using battery power. It will utilize a keypad to take inputs from the conductor and outputs will be delivered to the monochrome LCD display, thermal printer and a buzzer as a usual ticketing machine. Crowd

information will be recorded and reported to the server by tracking the issued tickets count and start and destination of the issued tickets. The GPRS (General Packet Radio Service) module will allow the device to access cellular networks using a SIM card thus enabling HTTP/IP communication between the webserver and device. The GPS receiver module will capture location/ velocity and time data from the GPS/GNSS networks, and captured data will be continuously delivered to the webserver. This data will be processed and displayed on the commuter mobile app and admin web app so that end users can see next arriving bus, estimated times, number of busses in the route, occupancy of the bus, and other additional features. Also drivers and conductors will be able to see the number of waiting commuters through the device as the server and IoT device communication is bi-directional. The IMU sensor will be used to detect rapid acceleration/decelerations and that data will be processed along with location data real-time, in the server or on-device, using a pre-trained ML model, to identify irregular, unsafe driving patterns and crash incidents.

3.2) User Scenario

For conductors, the device will function as a standard ticketing machine, requiring no extra effort. For commuters, the app will notify when a bus is ready to depart from the depot, displaying its real-time location and estimated arrival times at stops. This allows users to adjust their schedules and minimize waiting times. Additionally, real-time occupancy data will help commuters decide whether to board the current bus, wait for the next, or take an alternate route. Commuters can also mark themselves as “waiting” for routes with flexible or night schedules. This lets drivers see the number of waiting passengers and adjust their routes slightly to improve convenience and safety.

When scaled across the entire network, the system can analyze collected data to optimize routes, schedules, and resource allocation, making the transportation system more efficient and beneficial for all parties involved.

Team Details



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4) Additional Information

For the system to be successfully implemented and deliver its full benefits, it must be deployed across all buses within at least one urban area. Additionally, to ensure the system's effectiveness, all commuters must be required by law to obtain tickets.

To accommodate all demographics, the **e-ticketing system** should be introduced alongside traditional paper tickets, as some passengers may lack the technological literacy to adopt digital payments immediately. Furthermore, while e-ticketing is a crucial enhancement, it may not be included in the **Minimum Viable Product (MVP)** and can be introduced as a future extension based on the system's success.